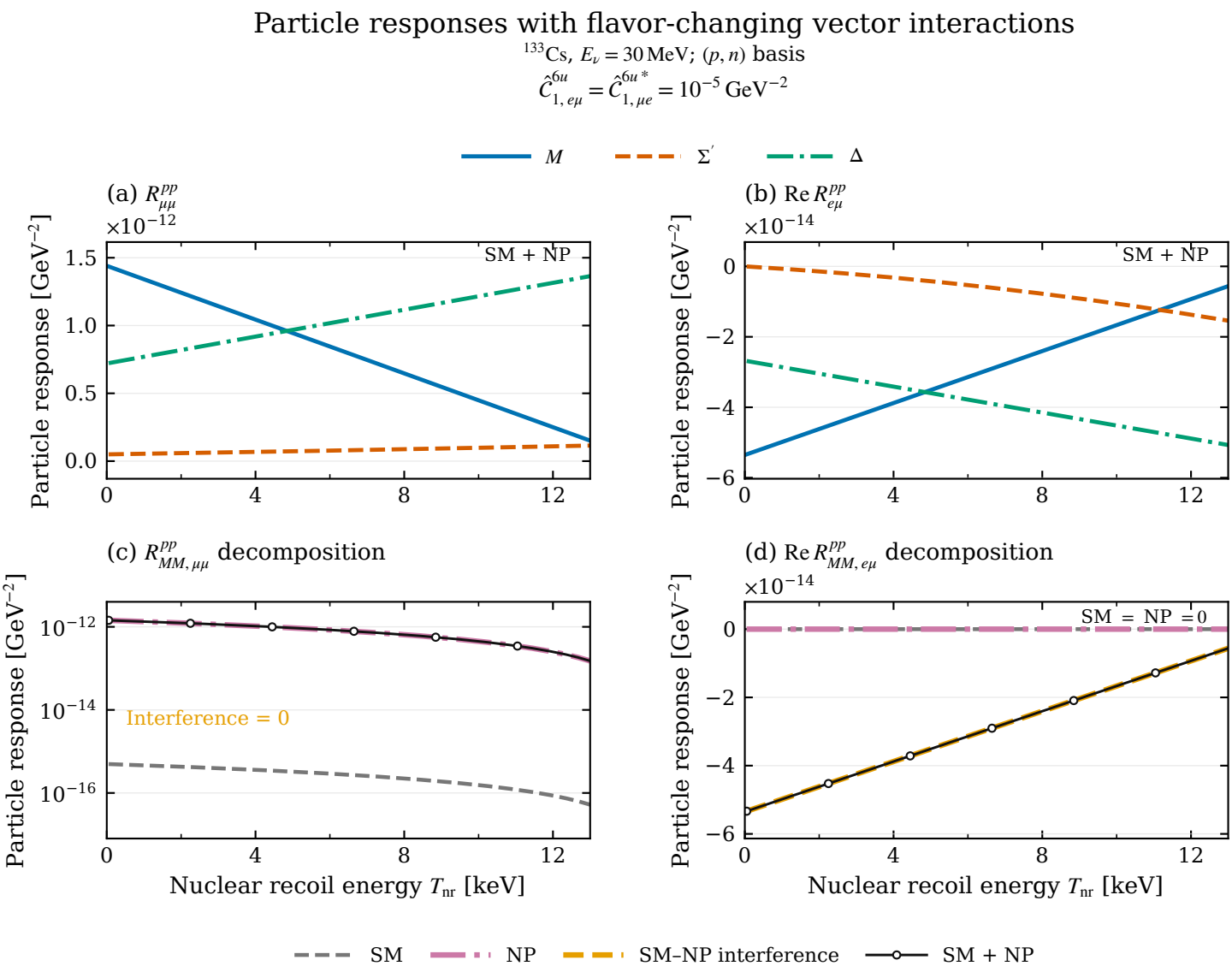

Particle responses with flavor-changing vector interactions
^{133}Cs, $E_\nu = 30$ MeV; (p, n) basis
$\hat{\mathcal{C}}^{6u}_{1,e\mu} = \hat{\mathcal{C}}^{6u\,*}_{1,\mu e} = 10^{-5}$ GeV^{-2}
M
Σ'
Δ
(a) $R^{pp}_{\mu\mu}$
$\times 10^{-12}$
SM + NP
Particle response [GeV^{-2}]
(b) Re $R^{pp}_{e\mu}$
$\times 10^{-14}$
SM + NP
Particle response [GeV^{-2}]
(c) $R^{pp}_{MM,\mu\mu}$ decomposition
Interference = 0
Particle response [GeV^{-2}]
Nuclear recoil energy $T_{\rm nr}$ [keV]
(d) Re $R^{pp}_{MM,e\mu}$ decomposition
$\times 10^{-14}$
SM = NP = 0
Particle response [GeV^{-2}]
Nuclear recoil energy $T_{\rm nr}$ [keV]
SM
NP
SM–NP interference
SM + NP